LEARNER GUIDE

# Circumference of a Circle

2D and 3D Shapes. A short guide to walk beside you — read as much or as little as helps.
*Connection before curriculum. Always.*

## What this lesson is about

The distance around a circle is called its circumference. In this lesson you discover something remarkable: the circumference is always the same number of times the diameter — a little more than three — for every circle, large or small. That number is π (pi). The question you are exploring is: how far is it around a circle, and how does that distance depend on the circle's width?

## What you are learning to notice

- The parts of a circle: the centre, the radius, the diameter (d = 2r) and the circumference.
- That the circumference divided by the diameter is always about 3.14 — the number π.
- That the circumference is π times the diameter: C = π d (and C = 2 π r).
- That the points (d, C) lie on a straight line through the origin, whose gradient is π.

## Moving through it, one step at a time

Use the (d, C) Plotter: change the diameter, plot the point G = (d, C), and watch the points line up. Try to find the gradient and write the equation yourself before you reveal them. Roll the wheel one full turn to see the circumference stretch out to about three diameters. Then estimate the distance around real objects. There is no rush: predict before you check, and use the symbol palette to enter π.

> **A gentle reminder**
> You can pause. You can return. You do not need to complete everything in one sitting.

## If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (multiplying by a decimal, ratio (‘times bigger’), and the names of the parts of a circle), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

**How the worked solution unlocks**

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

## Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

## Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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